National University of Singapore
School of Computing
CS3244 Machine Learning

Tutorial 4: Neural Networks

Issue: February 21, 2020 Due: March 2, 2020 (10 am)

Important Instructions:

- *Your solutions for this tutorial must be TYPE-WRITTEN.*
- *SUBMIT YOUR SOLUTIONS in PDF format to the 'TUTORIAL 4 SUBMISSION' folder under Files in LumiNUS by the DUE DATE specified above. Late submissions will NOT be entertained.*
- *Indicate your NAME, STUDENT NUMBER, and TUTORIAL GROUP in your submitted solution.*
- *YOUR SOLUTION TO QUESTION* **BL 7** *will be GRADED for this tutorial.*
- *You may discuss the content of the questions with your classmates. But everyone should work out and write up ALL the solutions by yourself.*

BL 6 Supposing the weights w_1 and w_2 of a perceptron (see page 6 of "Neural Networks" lecture slides) are both set to the value of **0.5**, derive the **largest** possible range of the values of w_0 that can be set for the perceptron to represent the AND gate (i.e., AND(x_1, x_2)). Assume that the inputs x_1 and x_2 and output $o(x_1, x_2)$ of the perceptron are Boolean with the values of **1 or –1**.
Show the steps of your derivation. **No marks will be awarded for not doing so.**

Solution. $-1 < w_0 \leq 0$.

BL 7 (Final Exam AY2017/18) Supposing the weights w_1 and w_2 of a perceptron (see page 6 of "Neural Networks" lecture slides) are both set to the value of **–1**, derive the **largest** possible range of the values of w_0 that can be set for the perceptron to represent the NAND gate (i.e., NAND(x_1, x_2)). Assume that the inputs x_1 and x_2 and output $o(x_1, x_2)$ of the perceptron are Boolean with the values of **1 or –1**.
Show the steps of your derivation. **No marks will be awarded for not doing so.**

Solution. $0 < w_0 \leq 2$.

BL 8 In every multi-layer network of perceptron units described below,

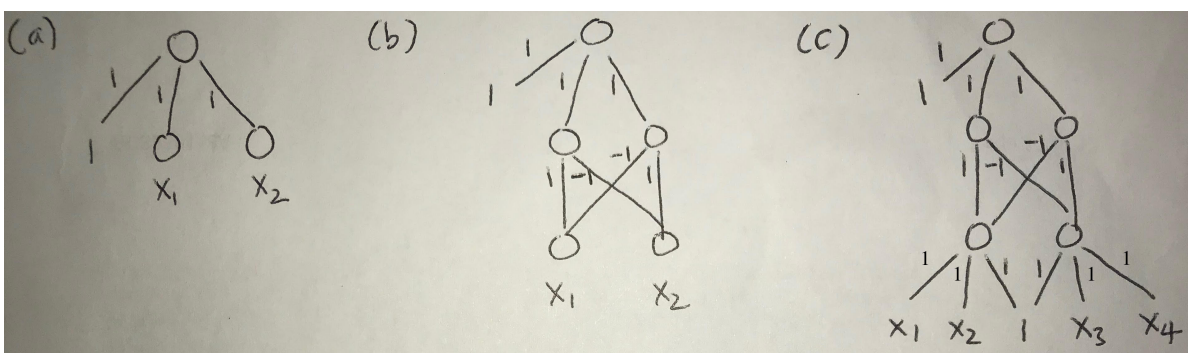
- There should be only one (Boolean) output node and an input node for every (Boolean) input.
- A Boolean is -1 if false, and 1 if true.
- The activation function of every (non-input) node is a -1 to 1 step function, including that of the output node (see page 6 of the "Neural Networks" lecture slides).
- Your weights must be **integers** and kept **small**, but possibly negative.
- Keep your networks as symmetric as possible – doing this in questions a and b may help you in question c.
- You don't have to draw edges with weight 0.

(a) Construct and draw a perceptron network with no hidden layers that implements (x_1 OR x_2).

(b) Construct and draw a perceptron network with one hidden layer (with two nodes) that implements (x_1 XOR x_2).

(c) Combining your constructions in questions a and b, construct and draw a perceptron network with two hidden layers (two nodes each) that implements (x_1 OR x_2) XOR (x_3 OR x_4).

Solution.



TM 4.1 What are the values of weights w_0, w_1, and w_2 for the perceptron whose decision surface is illustrated in figure a on page 7 of the "Neural Networks" lecture slides? Assume the surface crosses the x_1 axis at -1, and the x_2 axis at 2.

Solution. $w_0 = w_1 = -2w_2$, $w_0, w_1 < 0$ and $w_2 > 0$. For example, $w_0 = -2$, $w_1 = -2$, and $w_2 = 1$.

TM 4.3 Consider two perceptrons defined by the threshold expression $w_0 + w_1x_1 + w_2x_2 > 0$. Perceptron A has weight values $w_0 = 1$, $w_1 = 2$, $w_2 = 1$ and perceptron B has the weight values $w_0 = 0$, $w_1 = 2$, $w_2 = 1$. True or false? Perceptron A is *more general than* perceptron B; the notion of *more general than* is previously defined on page 12 of the "Concept Learning" lecture slides.

Solution. True.